Wave Data Preparation: Copernicus/EMODnet In-Situ Sea-State Data for WEC Power Estimation

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Source notebook: `01_wave_data_preparation.ipynb`

01 — Wave Data Preparation: Copernicus/EMODnet In-Situ Sea-State Data for WEC Power Estimation

1. Purpose

This notebook retrieves, inspects, cleans, and prepares measured in-situ wave observations from the Leixões coastal buoy.

The prepared dataset provides the sea-state time series needed for the next analyses, including simplified WEC power estimation from wave-height and period variables, short-term forecasting, prediction intervals, and storage-aware smoothing metrics.

2. Data source

The selected source is the Leixões coastal buoy, available through Copernicus Marine / EMODnet Physics in-situ wave observations.

Useful source pages:

- EMODnet Map Viewer
- Leixões platform page on EMODnet Physics
- Copernicus Marine product page: Global Ocean - Delayed Mode Wave product
- Copernicus Marine Toolbox documentation
- Copernicus Marine product user manual
- Copernicus Marine quality information document

Data citation:

> Global Ocean - Delayed Mode Wave product. E.U. Copernicus Marine Service Information (CMEMS). Marine Data Store (MDS). DOI: 10.17882/70345. Accessed 24 Jun 2026.

3. Selected location

The selected platform is the Leixões coastal buoy, located off northern Portugal near the Atlantic coast around Porto.

Portugal has documented ocean-energy activity along its Atlantic coast, including wave-energy testing and demonstration areas such as Aguçadoura and Viana do Castelo. The Leixões buoy is used here as an open measured wave-data source from this broader Atlantic-facing region.

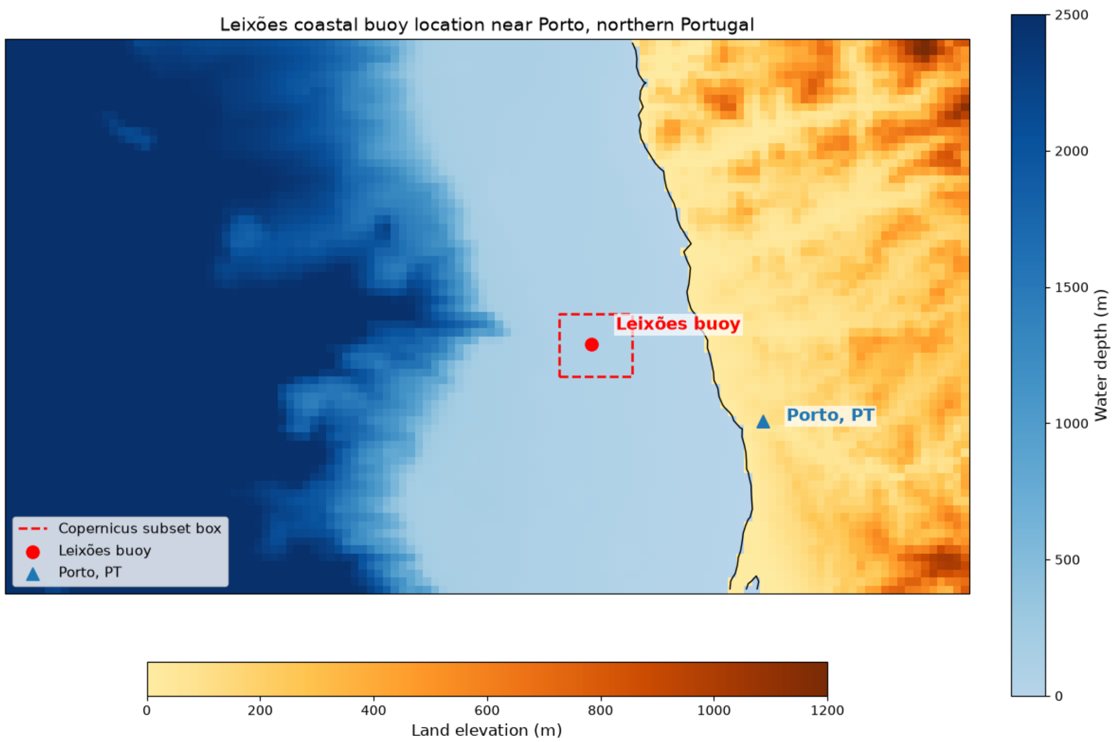
General context reference:

- IEA-OES Portugal country page

4. Notebook outputs

The notebook prepares:

- a raw NetCDF file downloaded from Copernicus Marine,
- a processed modelling-ready time series for the following notebooks.



`Saved figure to: ../outputs/figures/study_location_leixoes_coastal_buoy.png`

Map of the study location near the Leixões coastal buoy off northern Portugal. Bathymetry and land elevation are derived from NOAA ETOPO 2022 topography/bathymetry data. The dashed box indicates the spatial subset used later to retrieve buoy observations from Copernicus Marine.

Map background data citation:

> NOAA National Centers for Environmental Information. 2022: ETOPO 2022 15 Arc-Second Global Relief Model. NOAA National Centers for Environmental Information. DOI: 10.25921/fd45-gt74. Accessed 24 Jun 2026.

5. Data product and variables

The working Copernicus Marine dataset and platform used in this notebook are:

Item	Value
Product	`INSITU_GLO_WAV_DISCRETE_MY_013_045`
Dataset ID	`cmems_obs-ins_glo_wav_my_na_irr`
Platform	Leixões coastal buoy
Platform code	`Leixoes-coast-buoy`
Platform type	Mooring
Institution / provider	Hydrographic Institute
Approximate latitude	`41.317`
Approximate longitude	`-8.983`

Item	Value
Native sampling observed during testing	Approximately 30 minutes

The variables requested from Copernicus Marine are:

Variable	Meaning	Unit	Use in this notebook and later analysis
`time`	Observation timestamp	UTC datetime	Main time index for wave-data preparation.
`latitude`	Platform latitude	degrees north	Confirms station location.
`longitude`	Platform longitude	degrees east	Confirms station location.
`VHM0`	Spectral significant wave height, `Hm0`	m	Main wave-height variable for later simplified WEC power estimation and forecasting.
`VTPK`	Wave period at spectral peak / peak period, `Tp`	s	Main wave-period descriptor for later simplified power-matrix-style analysis.
`VTM02`	Spectral moments `(0,2)` wave period, `Tm02`	s	Additional period descriptor.
`VTZA`	Average zero-crossing wave period, `Tz`	s	Secondary period descriptor and quality/context variable.
`VTMX`	Maximum wave period, `Tmax`	s	Extreme/sea-state context variable.
`VTZM`	Period of the highest wave, `Thmax`	s	Extreme-wave context variable.
`VAVH`	Average height of highest one-third waves, `H1/3`	m	Secondary wave-height descriptor; useful for comparison with `VHM0`.
`VZMX`	Maximum zero-crossing wave height, `Hmax`	m	Extreme-wave context and quality-control support.

Variable	Meaning	Unit	Use in this notebook and later analysis
`VPED`	Wave principal direction at spectral peak	degrees	Directional feature for site characterization and optional forecasting features.
`VPSP`	Wave directional spreading at spectral peak	degrees	Directional spreading feature for site characterization.
`*_QC`	Quality-control flags for corresponding variables	code	Used to inspect and filter data quality where appropriate.

6. Retrieving raw Copernicus Marine wave observations

The following cell downloads the selected Leixões buoy wave variables from the Copernicus Marine in-situ wave dataset and saves the raw NetCDF file under `../data/raw/`.

The requested variable list includes the wave-height, period, directional, and extreme-wave descriptors confirmed during testing. `VTM10` is also requested here to check whether the spectral moments `(-1,0)` period is available for this platform.

Copernicus Marine login

Downloading data with the Copernicus Marine Toolbox requires a Copernicus Marine account.

Registration page:

- Copernicus Marine registration

After registration, log in once:

```
Raw file already exists, skipping download:
../data/raw/copernicus_leixoes_wave_20250202_20260624_raw/Leixoes-coast-buoy.nc
```

7. Raw data retrieval summary

The downloaded NetCDF file is inspected below to confirm the actual time coverage, platform metadata, retrieved variables, missingness, and whether all requested variables were available for this platform.

Item	Value
Raw file	../data/raw/copernicus_leixoes_wave_20250202_20260624_raw/Leixoes-coast-buoy.nc
File size (MB)	1.24
Time records	6223
First timestamp	2025-02-02 00:56:25
Last timestamp	2025-06-30 23:32:12

Item	Value
Median sampling interval (min)	30.0
Most common sampling interval (min)	30.0

Item	Value
Platform code	Leixoes-coast-buoy
Platform type	MO
Institution	Hydrographic Institute
Dataset ID	cmems_obs-ins_glo_wav_my_na_irr
Latitude	41.316666
Longitude	-8.983334

Variable	Available	Standard name	Units	Missing (%)	Ancillary variable
VHM0	yes	sea surface wave significant height	m	0.02	VHM0_QC
VTPK	yes	sea surface wave period at variance spectral density maximum	s	0	VTPK_QC
VTM10	no			nan	
VTM02	yes	sea surface wave mean period from variance spectral density second frequency moment	s	0	VTM02_QC
VTZA	yes	sea surface wave mean period	s	1.38	VTZA_QC
VTMX	yes	sea surface wave maximum period	s	1.38	VTMX_QC
VTZM	yes	sea surface wave period of highest wave	s	1.38	VTZM_QC
VAVH	yes	sea surface wave significant height	m	1.38	VAVH_QC
VZMX	yes	sea surface wave maximum height	m	1.38	VZMX_QC
VPED	yes	sea surface wave from direction at variance spectral density maximum	degree	0	VPED_QC
VPSP	yes	sea surface wave directional spread at variance spectral density maximum	degree	0	VPSP_QC

Although the Copernicus Marine request window was set from February 2025 to June 2026, the retrieved delayed-mode file contains available records for February–June 2025. This actual retrieved coverage is used for the following short-term forecasting, prediction-interval, and storage-smoothing examples.

The record has a typical 30-minute sampling interval and is not treated as a full annual dataset. Later analyses therefore do not present annual or seasonal performance conclusions.

`VTM10` was requested as an energy-period-related descriptor for later WEC power estimation, but it was not returned for this platform/file. The following WEC estimation step will therefore use the available period descriptors, mainly `VTPK` and optionally `VTM02` / `VTZA`, with the chosen period variable stated there.

8. Quality and sampling intervals checks

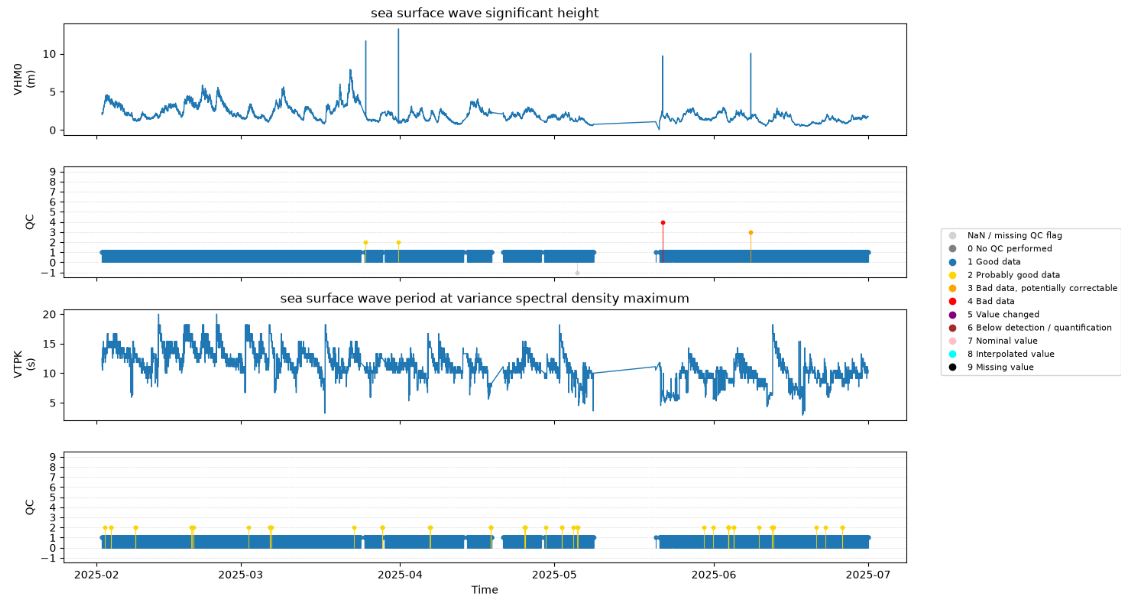
This section:

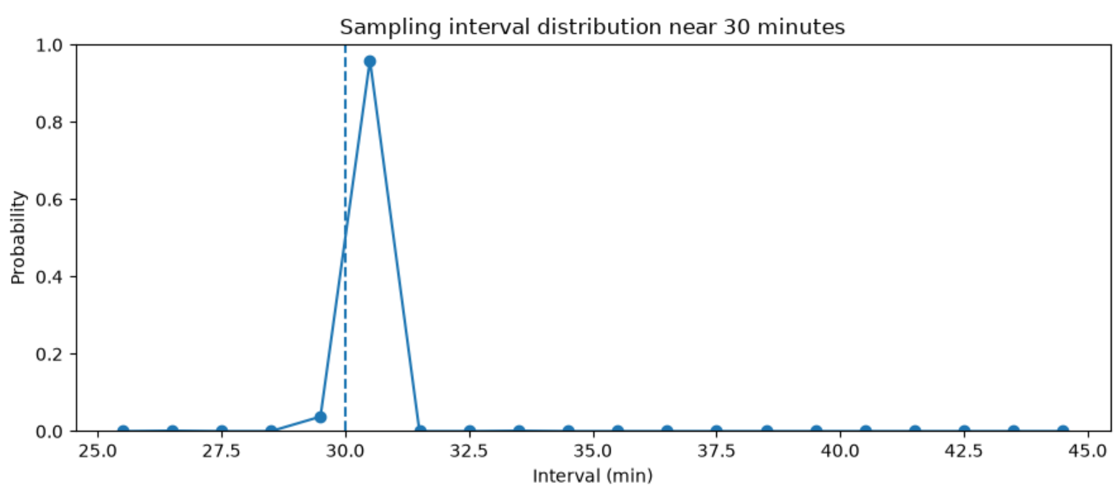
- converts the NetCDF file into a flat table,
- keeps retrieved wave variables and their dataset-declared QC variables,
- lists only QC flags different from `1` / good data,
- plots key wave variables with aligned QC/missingness strips,
- checks whether sampling intervals are concentrated around the 30-minute grid.

Copernicus In Situ QC flag meanings follow the reference table in the Copernicus In Situ TAC documentation:

Flag	Meaning
0	No QC performed
1	Good data
2	Probably good data
3	Bad data, potentially correctable
4	Bad data
5	Value changed
6	Below detection / quantification
7	Nominal value
8	Interpolated value
9	Missing value

Variable	QC variable	QC flag	Meaning	Count
VHM0	VHM0_QC	2	2 Probably good data	2
VHM0	VHM0_QC	3	3 Bad data, potentially correctable	1
VHM0	VHM0_QC	4	4 Bad data	1
VHM0	VHM0_QC	nan	Missing QC flag	1
VTPK	VTPK_QC	2	2 Probably good data	48
VTM02	VTM02_QC	2	2 Probably good data	2
VTZA	VTZA_QC	nan	Missing QC flag	86
VTMX	VTMX_QC	2	2 Probably good data	9
VTMX	VTMX_QC	nan	Missing QC flag	86
VTZM	VTZM_QC	2	2 Probably good data	2
VTZM	VTZM_QC	nan	Missing QC flag	86
VAVH	VAVH_QC	nan	Missing QC flag	86
VZMX	VZMX_QC	nan	Missing QC flag	86





The retrieved record is mostly sampled at 30-minute intervals, with one visible data gap around May 2025. The processed table therefore regularizes the data to a 30-minute grid while keeping missing periods as `NaN`.

Most QC flags indicate good data. A small number of non-good or missing QC flags are present, mainly for `VHM0`, `VTPK`, and selected secondary variables.

9. Processing and saving data

This section:

- keeps values with QC flags `1` / good and `2` / probably good,

- sets other flagged values to missing,
- renames columns for clearer downstream use,
- aligns observations to a regular 30-minute grid without interpolation,
- saves the processed table to `../data/processed/leixoes_wave_30min_processed.parquet`.

Column renaming used for the processed table:

Raw variable	Processed name	Meaning
`VHM0`	`hm0_m`	Spectral significant wave height
`VTPK`	`tp_s`	Peak wave period
`VTM02`	`tm02_s`	Spectral moments `(0,2)` wave period
`VTZA`	`tz_s`	Average zero-crossing wave period
`VTMX`	`tmax_s`	Maximum wave period
`VTZM`	`thmax_s`	Period of the highest wave
`VAVH`	`h13_m`	Average height of highest one-third waves
`VZMX`	`hmax_m`	Maximum zero-crossing wave height
`VPED`	`dir_peak_deg`	Wave direction at spectral peak
`VPSP`	`spread_peak_deg`	Directional spreading at spectral peak

```
                 time   latitude  longitude  hm0_m  tp_s  tm02_s  tz_s  \
0 2025-02-02 01:00:00  41.316666  -8.983334   2.24  12.5     8.2   8.7
1 2025-02-02 01:30:00  41.316666  -8.983334   2.00  11.8     8.0   8.3
2 2025-02-02 02:00:00  41.316666  -8.983334   2.00  11.1     7.4   8.1
3 2025-02-02 02:30:00  41.316666  -8.983334   2.05  11.1     7.3   7.7
4 2025-02-02 03:00:00  41.316666  -8.983334   2.19  11.8     7.8   8.1

   tmax_s  thmax_s  h13_m  hmax_m  dir_peak_deg  spread_peak_deg
0    16.4     10.9   2.11    3.21         315.0             16.0
1    15.6     12.5   1.83    2.93         301.0             19.0
2    16.4      9.4   1.88    3.46         304.0             22.0
3    16.4      9.4   1.86    2.72         305.0             24.0
4    15.6     10.2   1.99    2.92         302.0             18.0
```

Item	Value
Processed rows	7150
First timestamp	2025-02-02 01:00:00
Last timestamp	2025-06-30 23:30:00
Columns	time, latitude, longitude, hm0__m, tp__s, tm02__s, tz__s, tmax__s, thmax__s, h13__m, hmax__m, dir__peak__deg, spread__peak__deg
Processed output	../data/processed/leixoes__wave__30min__processed.parquet


```
Saved comparison figure to:
../outputs/figures/raw_qc_processed_comparison_leixoes_coastal_buoy.png
```